Short-Kappa Maximal-Parabolic Induction

Dehydrated Proof

Condensed from the Rethlas-verified blueprint

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Let $F=\mathbb{R}$ and $\epsilon=\pm1$. Let $(V_0,\langle\ ,\ \rangle_0)$ be non-degenerate ϵ-symmetric, let $G_0=\mathrm{Isom}(V_0)$, and let $X_0\in\mathfrak{g}_0$ be nilpotent. Put

$$c_1=\dim\ker X_0,\qquad c_2=\dim(\ker X_0\cap\operatorname{Im}X_0)=\dim(\ker X_0^2/\ker X_0).$$

Let E,E' be κ-dimensional real vector spaces with perfect pairing $\lambda:E\times E'\to\mathbb{R}$, and assume

$$c_2\le\kappa<c_1,\qquad d:=c_1-\kappa.$$

On $V=E\oplus V_0\oplus E'$ use

$$\langle e+v+e',f+w+f'\rangle=\lambda(e,f')+\epsilon\lambda(f,e')+\langle v,w\rangle_0.$$

Let $G=\mathrm{Isom}(V)$, let $P\subset G$ stabilize E, and let

$$\mathfrak{u}=\{Y\in\mathfrak{g}:Y(E)=0,\ Y(E^\perp)\subset E,\ Y(V)\subset E^\perp\}.$$

We compute

$$\mathrm{Ind}_P^G(\mathcal{O}_0)=\overline{G\cdot(\mathcal{O}_0+\mathfrak{u})},\qquad\mathcal{O}_0=G_0\cdot X_0,$$

with real-topological closure. Since X_0 is skew-adjoint,

$$(\operatorname{Im}X_0)^\perp=\ker X_0.$$

Thus the form on $\ker X_0$ has radical $R:=\ker X_0\cap\operatorname{Im}X_0$, and descends to a non-degenerate ϵ-symmetric form on

$$K_{X_0}:=\ker X_0/R.$$

Let $\mathscr{A}=\mathrm{Gr}_d(K_{X_0})$. Let $\mathscr{A}^{\mathrm{mr}}\subset\mathscr{A}$ be the set of d-planes on which the restricted form has maximal possible rank.

Theorem. *The maximal G-orbits in $\mathrm{Ind}_P^G(\mathcal{O}_0)$ are parametrized by the isometry classes, inside K_{X_0}, of subspaces $A\in\mathscr{A}^{\mathrm{mr}}$. For the orbit $\mathcal{O}_A=G\cdot x_A$ attached to A,*

$$m(\mathcal{O}_A,P)=\frac{|\pi_0(Z_G(x_A))|}{|\pi_0(Z_G(x_A)\cap P)|}=\begin{cases}1, & \epsilon=+1,\\ 1, & \epsilon=-1\text{ and }d\text{ is even},\\ 2, & \epsilon=-1\text{ and }d\text{ is odd}.\end{cases}$$

Here π_0 means connected components for the usual real topology.

Proof. First fix a non-degenerate lift $V_1\subset\ker X_0$ of K_{X_0}, so that $\ker X_0=V_1\oplus R$ and $V_1\simeq K_{X_0}$. Every element of $X_0+\mathfrak{u}$ is uniquely

$$X_{C,B}=\begin{pmatrix}0&C^\vee&B\\0&X_0&C\\0&0&0\end{pmatrix},\qquad C:E'\to V_0,\quad B:E'\to E,$$

where

$$\lambda(C^\vee v,e')=-\langle v,Ce'\rangle_0,\qquad\lambda(Bu,v)+\epsilon\lambda(Bv,u)=0.$$

For $D:E'\to V_0$, the element

$$u_D=\begin{pmatrix}1&D^\vee&\frac12 D^\vee D\\0&1&D\\0&0&1\end{pmatrix}\in P$$

satisfies

$$u_DX_{C,B}u_D^{-1}=X_{C-X_0D,\ B+D^\vee C-C^\vee D-D^\vee X_0D}.\tag{1}$$

The open part is best read directly from the C-block. Put

$$\phi_C:=C^\vee|_{\ker X_0}:\ker X_0\to E.$$

The map $C\mapsto\phi_C$ from $\mathrm{Hom}(E',V_0)$ to $\mathrm{Hom}(\ker X_0,E)$ is surjective: modulo $\operatorname{Im}X_0$, it is exactly the identification

$$\mathrm{Hom}(E',V_0/\operatorname{Im}X_0)\simeq\mathrm{Hom}(\ker X_0,E)$$

coming from the two perfect pairings. The first rank-open condition is ϕ_C surjective. Then

$$L_C:=\ker\phi_C$$

has dimension $c_1 - \kappa = d$. The second rank-open condition is $\phi_C|_R : R \to E$ injective, equivalently $L_C \cap R = 0$. Hence the quotient map $q : \ker X_0 \to K_{X_0}$ sends L_C isomorphically onto a d-plane

$$A_C := q(L_C) \subset K_{X_0}.$$

The last rank-open condition is $A_C \in \mathscr{A}^{\mathrm{mr}}$. Thus the desired open subset is

$$\Omega = \{X_{C,B} : \phi_C \text{ is surjective},\ \phi_C|_R \text{ is injective},\ A_C \in \mathscr{A}^{\mathrm{mr}}\}. \tag{2}$$

To see this is a nonempty polynomial open set, choose $V_+ \subset V_0$ with

$$V_0 = V_+ \oplus \operatorname{Im} X_0, \qquad V_+ \cap \ker X_0 = V_1.$$

For $C : E' \to V_0$, write $S_C : E' \to V_+$ for its quotient modulo $\operatorname{Im} X_0$. Then ϕ_C is the transpose of S_C, so

$$\phi_C \text{ surjective} \iff S_C \text{ injective},$$

and $\phi_C|_R$ is injective iff

$$\overline{S}_C : E' \to V_+/(V_+ \cap \ker X_0)$$

is surjective. Thus the first generic locus is represented by injective maps $S : E' \to V_+$. The further condition $q(\ker \phi_C) \in \mathscr{A}^{\mathrm{mr}}$ can be read as maximal possible rank of the Gram matrix

$$(u, v) \mapsto \langle S(u), S(v)\rangle_0$$

among such S with $\overline{S}$ surjective. These are exactly nonvanishing-minor conditions, and they are nonempty by choosing S with maximal possible Gram rank.

Now define the normal forms only after A has appeared from the open conditions. For $A \in \mathrm{Gr}_d(K_{X_0})$, let $W_A \subset V_0/\operatorname{Im} X_0$ be the annihilator of A under $(V_0/\operatorname{Im} X_0) \times \ker X_0 \to \mathbb{R}$. Then $\dim W_A = \kappa$. Choose a lift $V_A \subset V_0$ mapping isomorphically to W_A, choose $S_A : E' \xrightarrow{\sim} V_A$, and put

$$x_A := X_{S_A,0}.$$

The defining annihilator property gives

$$\ker(S_A^\vee|_{\ker X_0}) = A, \qquad S_A^\vee|_{\ker X_0} : \ker X_0 \to E \text{ is surjective}. \tag{3}$$

Take $x = X_{C,B} \in \Omega$, and put $A = A_C$. A Levi conjugation first makes $\phi_C = S_A^\vee|_{\ker X_0}$. Then $C - S_A$ has image in $\operatorname{Im} X_0$, so (1) makes $C = S_A$. A second conjugation by some u_D with $D(E') \subset \ker X_0$ kills the remaining B-block. Thus every point of Ω is P-conjugate to x_{A_C}. Conversely each x_A with $A \in \mathscr{A}^{\mathrm{mr}}$ lies in Ω, so the normal-form conjugations give

$$\Omega = P \cdot \{x_A : A \in \mathscr{A}^{\mathrm{mr}}\}. \tag{4}$$

The same argument shows that x_A is independent, up to P-conjugacy, of the choices of V_A, S_A. If two subspaces $A, A' \subset K_{X_0}$ are related by an ambient isometry of K_{X_0}, that isometry lifts to an element of $Z_{G_0}(X_0)$, hence x_A and $x_{A'}$ are P-conjugate.

Moreover Ω is a finite disjoint union

$$\Omega = \bigsqcup_i P \cdot x_{A_i}, \tag{5}$$

where A_i run over the isometry classes in $\mathscr{A}^{\mathrm{mr}}$. These classes are: non-degenerate symmetric d-planes classified by signature if $\epsilon = +1$; one non-degenerate symplectic class if $\epsilon = -1$ and d is even; and one alternating rank $d-1$ class with a one-dimensional radical if $\epsilon = -1$ and d is odd. The ambient transitivity in each fixed class is Witt extension in the symmetric case and extension of symplectic bases in the alternating cases.

Set $Y = G \cdot (X_0 + \mathfrak{u})$. For $x_i = x_{A_i}$, the piece $P \cdot x_i$ is open in the affine slice, hence

$$T_{x_i}(X_0 + \mathfrak{u}) = T_{x_i}(P \cdot x_i) = [\mathfrak{p}, x_i] = \mathfrak{u}.$$

For the action map $G \times (X_0 + \mathfrak{u}) \to \mathfrak{g}$, the differential at $(1, x_i)$ has image

$$[\mathfrak{g}, x_i] + \mathfrak{u} = [\mathfrak{g}, x_i],$$

so $G \cdot x_i$ is open in Y. Since Ω is dense in the slice,

$$\operatorname{Ind}_P^G(\mathcal{O}_0) = \overline{Y} = \overline{G \cdot \Omega} = \bigcup_i \overline{G \cdot x_i}. \tag{6}$$

Thus every orbit in the induced set lies below one of the listed orbits. Because each $G \cdot x_i$ is open in Y, it has the dimension of $\overline{Y}$. If another orbit had $G \cdot x_i$ in its closure, the usual boundary dimension inequality for locally closed real algebraic orbits would force equality of the two orbits. Hence the $G \cdot x_i$ are exactly the maximal induced orbits.

They are distinct precisely as stated. For the normal form one computes

$$\ker x_A = E \oplus A, \qquad \operatorname{Im} x_A = E \oplus \operatorname{Im} X_0 \oplus S_A(E'), \qquad \ker x_A \cap \operatorname{Im} x_A = E \oplus \operatorname{rad}(A). \tag{7}$$

Therefore

$$\ker x_A/(\ker x_A \cap \operatorname{Im} x_A) \simeq A/\operatorname{rad}(A) \tag{8}$$

as bilinear spaces. This invariant recovers the signature in the symmetric case and gives no further splitting in the two alternating cases, exactly as the classification above says.

It remains to compute the multiplicity. If A is non-degenerate, then by (7)

$$\ker x_A = E \oplus A, \qquad \ker x_A \cap \operatorname{Im} x_A = E.$$

Since E is the radical of the bilinear space $\ker x_A$, every element of $Z_G(x_A)$ preserves E. Hence $Z_G(x_A) \subset P$, and

$$m(G \cdot x_A, P) = 1.$$

This covers $\epsilon = +1$ and $\epsilon = -1$ with d even.

Now assume $\epsilon = -1$ and d is odd. Write

$$A = A_0 \oplus \mathbb{R}r, \qquad \mathbb{R}r = \operatorname{rad}(A),$$

with A_0 symplectic, choose $s \in K_{X_0}$ with $\langle r, s\rangle_K = 1$, and choose $e \in E, e' \in E'$ with $\lambda(e, e') = 1$. Replacing x_A by a P-conjugate representative, we may arrange that

$$U = \mathbb{R}e \oplus \mathbb{R}r \oplus \mathbb{R}s \oplus \mathbb{R}e'$$

is stable and

$$x_A(e) = x_A(r) = 0, \qquad x_A(s) = e, \qquad x_A(e') = r. \tag{9}$$

On $W = U^\perp$, the kernel quotient is A_0, so the non-degenerate case gives $Z_{G(W)}(x_A|_W) \subset P(W)$.

In the ordered basis e, r, s, e',

$$x_A|_U = \begin{pmatrix} 0 & I_2 \\ 0 & 0 \end{pmatrix}, \qquad J_U = \begin{pmatrix} 0 & S_2 \\ -S_2 & 0 \end{pmatrix}, \qquad S_2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}.$$

Writing $g = \left(\begin{smallmatrix} A & B \\ C & D \end{smallmatrix}\right)$, the equations $gx_A = x_A g$ and $g^T J_U g = J_U$ give

$$C = 0, \qquad D = A, \qquad A^T S_2 A = S_2, \qquad M^T S_2 = S_2 M \quad (M = A^{-1}B).$$

Thus projection to A identifies $\pi_0(Z_{G(U)}(x_A|_U))$ with $\pi_0(O(S_2))$. Over $\mathbb{R}$,

$$O(S_2) = \left\{ \begin{pmatrix} a & 0 \\ 0 & a^{-1} \end{pmatrix} : a \in \mathbb{R}^\times \right\} \sqcup \left\{ \begin{pmatrix} 0 & a \\ a^{-1} & 0 \end{pmatrix} : a \in \mathbb{R}^\times \right\}.$$

The two diagonal branches and the two off-diagonal branches split further by $\operatorname{sgn}(a)$, so $|\pi_0(O(S_2))| = 4$. Intersecting with $P(U)$, i.e. imposing preservation of $\mathbb{R}e$, leaves only the diagonal branch, hence two real components. Therefore

$$|\pi_0(Z_{G(U)}(x_A|_U))| = 4, \qquad |\pi_0(Z_{G(U)}(x_A|_U) \cap P(U))| = 2. \tag{10}$$

Finally, relative to $V = U \oplus W$, block projection from $Z_G(x_A)$ to the product of the U- and W-centralizers, and the analogous projection after intersecting with P, are surjective with connected fibers. The same W-factor occurs in numerator and denominator because $Z_{G(W)}(x_A|_W) \subset P(W)$. By (10),

$$m(G \cdot x_A, P) = 4/2 = 2.$$

This completes the proof. $\square$